G. Recurrence With Square

time limit per test: 1 second

memory limit per test: 256 megabytes

The formula for Fibonacci numbers $F_i = 1 \cdot F_{i-1} + 1 \cdot F_{i-2}$ is an example of a linear recurrence relation. Let's consider a more general and difficult formula, which additionally involves the current index i.

Consider an infinite sequence defined by its first n elements $a_0, a_1, \ldots, a_{n-1}$ and coefficients $c_1, \ldots, c_n, p, q, r$. For every $i \geq n$:

$$a_i = (c_1 \cdot a_{i-1} + c_2 \cdot a_{i-2} + \cdots + c_n \cdot a_{i-n}) + p + i \cdot q + i^2 \cdot r$$

Find the value $a_k \bmod 10^9 + 7$.

Input

The first line contains two integers n $(1 \leq n \leq 10)$ and k $(1 \leq k \leq 10^{18})$.

The second line contains n integers $a_0, a_1, \ldots, a_{n-1}$ $(0 \leq a_i < 10^9 + 7)$ — the first n elements of the sequence.

The third line contains n integers $c_1, c_2, \ldots, c_n$ $(0 \leq c_i < 10^9 + 7)$.

The fourth and last line contains three integers p, q, r $(0 \leq p, q, r < 10^9 + 7)$.

Output

Print one line containing the number a_k modulo $10^9 + 7$.

Examples

input

```
2 2
0 30
2 1
2 1 1
```

output

```
68
```

input

```
1 3
5
1
2 3 4
```

output

```
85
```

Note

In the first sample, the sequence is defined as $a_0 = 0$, $a_1 = 30$, $a_i = (2a_{i-1} + a_{i-2}) + 2 + i + i^2$, so $a_k = a_2 = (2 \cdot 30 + 0) + 2 + 2 + 4 = 68$.

In the second sample, the first few elements of the sequence are $(5, 14, 38, 85, 163)$. As $k = 3$, we print $a_3 = 85$.

Matrix Exponentiation

Finished

Practice

→ Virtual participation

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Language: GNU G++23 14.2 (64 bit, ms

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336314598	Aug/31/2025 00:18	Happy New Year!